APRS COMMAND

A Free, Open-Source APRS Client for Amateur Radio

macOS · Windows 11 · Linux · Raspberry Pi

Version 1.1 · Updated July 27, 2026 · Applies to v0.5.0-alpha

APRS Command is a modern, cross-platform replacement for UI-View32 — the beloved APRS client whose author, Roger Barker G4IDE, passed away in 2004 with the source code destroyed per his wishes, permanently freezing it in time. APRS Command is released under GPL v3, written in C# and .NET 10, and runs identically on macOS, Windows 11, Linux, and Raspberry Pi from the same codebase, so it can never again be orphaned by a single developer or a single operating system.

Who It's For

Any licensed amateur radio operator who wants a live picture of APRS activity — at home, in the field, or running emergency communications. No prior APRS experience is required; APRS Command is equally at home guiding a first-time user through their first beacon and running an unattended digipeater/iGate station in a Pi-based field kit.

Core Capabilities

- Live interactive map of APRS stations with course/speed/altitude, symbols, trails, range rings, and a weather radar overlay.
- Four simultaneous connection paths: APRS-IS (internet), KISS-TCP (GrayWolf / Direwolf / any network KISS source), Serial KISS (hardware TNCs — Kantronics KPC-3+, Mobilinkd, DigiRig, TNC-Pi, Kenwood TM-D710), and AGWPE/BPQ32 (familiar to UI-View32 users).
- Managed Modem mode — APRS Command runs and configures Direwolf directly against a SignaLink or DigiRig audio interface, with no separate Direwolf setup required.
- Full messaging with delivery/ACK tracking, Net Control roster for EmComm nets, and standard ICS documentation export for after-action reporting.
- iGate and digipeater operation, weather packet decoding with NWS alert integration, and GPX/KML import/export with CalTopo/SARTopo live position forwarding.
- Packet Statistics Dashboard, five built-in Session Templates (Public Service, EmComm, Weekly Net, SOTA/POTA, Skywarn) plus custom templates, and OS-native Voice Readout.
- Mobile Companion — a lightweight, read-only web dashboard any phone on the same network can open to monitor the station remotely.

Getting Started

1. Install

Download the installer for your platform from github.com/KE4CON/APRS-Command/releases — a .dmg for macOS, a Setup.exe for Windows, or a .deb/.rpm package for Linux and Raspberry Pi. A portable zip/tar.gz is also available for anyone who prefers to run the app from a folder with no installation step. APRS Command is not code-signed (an open-source volunteer project doesn't carry a $300-500/year certificate), so every platform shows a one-time security warning on first launch — this is expected and is standard for unsigned open-source software.

2. First Launch

A short setup wizard asks for your callsign and location, then you're on the map. If you don't yet have an APRS-IS passcode, enter -1 to receive and view all APRS traffic immediately — you simply won't be able to transmit until you add your real passcode later. Find your passcode at apps.magicbug.co.uk/passcode.

3. Get on RF (Optional)

APRS-IS alone gets most people running within minutes over the internet. When you're ready to add radio, Settings › Connections supports as many ports as you need, of any type, active at once — plug in a hardware TNC, point APRS Command at a GrayWolf or Direwolf KISS-TCP source, or let APRS Command manage Direwolf itself against a SignaLink/DigiRig. The User Manual's RF and TNC Connection chapter walks through every path step by step, including exact baud rates and port-finding commands per platform.

Why APRS Command Exists

APRS was conceived by Bob Bruninga WB4APR not as a tracking system but as a situational-awareness tool — a shared, real-time picture of what's happening in an area: weather, resources, messages, and people. That vision drives every design decision in APRS Command. Being open source under GPL v3 means the project can never be discontinued or frozen the way UI-View32 was — any operator, anywhere, can pick it up and carry it forward.

Built for Field and Emergency Communications

APRS Command was designed from the start to work as well in a Pi-powered field kit running on battery as it does on a home desktop. A growing number of features exist specifically to support served-agency and EmComm operation:

- Net Control Roster — track check-ins, assignments, and status for an entire net in one window, built for real-time net operation.
- Session Templates — one-click configuration presets for Public Service events, EmComm activations, weekly nets, SOTA/POTA portable operation, and Skywarn spotting.
- After-Action Export — generate standard ICS-compatible documentation directly from a completed session.
- GrayWolf field-server pairing — GrayWolf (NW5W) is a single-binary Rust APRS station built for headless Raspberry Pi field deployment, combining an AFSK modem, digipeater, and iGate. APRS Command connects to it over KISS-TCP as the operator-facing client, so the radio-facing server can run unattended while the desktop client provides map, messaging, and roster tools.
- Mobile Companion — commanders and other operators on the same field network can pull up a live read-only dashboard on a phone without touching the primary operating position.
- Offline-friendly design — APRS-IS is optional; every core feature works over RF alone with no internet connection required.

A Note on Reliability

APRS Command's protocol layer is validated against a dedicated APRS Protocol 1.0.1 conformance test suite, and the APRS-IS ingestion path has been fuzz-tested against live regional traffic with zero crashes or misdecodes across every packet type the specification defines. The project favors verified, tested behavior over assumed correctness — a discipline that matters most exactly when the software is depended on in an emergency.

Supported Platforms

Platform	Minimum Requirement
macOS	macOS 12 Monterey or later. Apple Silicon (M1/M2/M3/M4) or Intel. 4 GB RAM.
Windows 11	Windows 10 or 11, 64-bit. 4 GB RAM.
Linux (64-bit)	Ubuntu 20.04+, Linux Mint 21+, Debian 11+, Fedora 36+, or any modern 64-bit distro. 4 GB RAM.
Raspberry Pi	Raspberry Pi 4 or 5. 4 GB RAM recommended. Raspberry Pi OS 64-bit (not 32-bit).

Status

Alpha — functional for daily use, under active development. Feedback and bug reports are welcome via GitHub Issues.

License

GNU General Public License v3. Free to use, study, modify, and redistribute — and permanently so, since GPL v3 prevents the project from ever being taken closed-source.

Where to Find Everything

- Source code and releases: github.com/KE4CON/APRS-Command
- User Manual — complete installation, setup, and operation guide covering every feature in depth.
- APRS Theory and Operations Primer — for anyone new to APRS: how packets, digipeaters, APRS-IS, and HF APRS actually work.
- Both manuals ship in the docs/ folder of the repository alongside this guide.

Built on the vision of Bob Bruninga WB4APR · In memory of Roger Barker G4IDE

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